

The Jacobian Conjecture in Dimension Two and the Role of the Condition $5''_s$

Łukasz Matysiak

Abstract

In this paper we analyze the endomorphic condition $5''_s$, introduced in two earlier publications of the author, and its consequences for Keller endomorphisms of the polynomial ring $A = k[x_1, \dots, x_n]$ over a field of characteristic 0. The condition $5''_s$ controls the behavior of square-free principal divisors in the images of endomorphisms and constitutes a key structural object in the study of the Jacobian Conjecture.

In the previous parts of the series it was shown that: (1) the ring A satisfies the condition $5'_s$, and therefore also the endomorphic condition $5''_s$; (2) Drużkowski-type endomorphisms $F(x) = x + (Bx)^3$ satisfy the condition $5''_s$ in every dimension; (3) if the image of a Keller endomorphism satisfies the condition $5''_s$, then the endomorphism is an automorphism (the JC- $5''_s$ theorem).

After Alpöge's announcement of a counterexample in dimension $n = 3$, it is known that there exist Keller endomorphisms which are not automorphisms. By the JC- $5''_s$ theorem, their images must violate the condition $5''_s$, which manifests itself through the disappearance of square-free principal divisors in factorizations of the form $\varphi(f) = gh$. In this paper we give a structural description of this mechanism of failure.

The main result of the paper is the proof that in dimension $n = 2$ a Keller endomorphism cannot violate the condition $5''_s$ in a way that leads to a counterexample. Combined with the classical dimension argument, this yields a complete proof of the Jacobian Conjecture in two variables. The proof appears in two independent forms: a classical dimension-theoretic argument and a structural criterion based on the condition $5''_s$. Thus dimension 2 turns out to be the unique case in which the Jacobian Conjecture remains true.

This paper constitutes the third part of a series devoted to factorizational criteria for invertibility of Keller endomorphisms and identifies the precise structural region in which the behavior of the condition $5_s''$ and the validity of JC(2) are determined.

1 Introduction

The Jacobian Conjecture (JC), formulated by O. Keller in 1939, is one of the most fundamental and at the same time most resistant problems in modern algebra. Its statement is deceptively simple: if an endomorphism

$$\varphi : k[x_1, \dots, x_n] \longrightarrow k[x_1, \dots, x_n]$$

over a field of characteristic 0 has Jacobian equal to a constant invertible unit, then φ should be an automorphism. Such endomorphisms are called Keller endomorphisms. Classical expositions of the Jacobian Conjecture can be found, for example, in [2, 5].

For more than eighty years JC remained open in all dimensions $n \geq 2$. The situation changed dramatically in 2026, when T. Alpöge constructed an explicit counterexample in dimension $n = 3$ [1]: there exists a Keller endomorphism

$$\Phi : k[x_1, x_2, x_3] \rightarrow k[x_1, x_2, x_3]$$

with constant nonzero Jacobian which is not an automorphism. By stabilization one immediately obtains that the Jacobian Conjecture is false in all dimensions $n \geq 3$. Thus the only unresolved case remains dimension $n = 2$, which is regarded as “critical”: on the one hand it does not admit many of the pathologies appearing in higher dimensions (cf. [2]), while on the other hand no proof or counterexample had been known.

In two earlier papers the author introduced the endomorphic condition $5_s''$ [8, 9], which controls the behavior of square-free principal divisors in the images of endomorphisms. This condition turned out to be extremely useful: it allows one to formulate the JC- $5_s''$ theorem, which states that if the image of a Keller endomorphism satisfies the condition $5_s''$, then the endomorphism is an automorphism (Theorem 9.2 in [8], Theorem 7.1 in [9]).

Twierdzenie 1.1 (Endomorphic version — The JC- $5_s''$ Theorem). *For endomorphisms with constant Jacobian the following conditions are equivalent:*

- *the endomorphism is an automorphism,*
- *its cubic form in the Druzkowski reduction satisfies the condition $5_s''$,*
- *the endomorphism preserves square-free elements with linear factors.*

In the first and second parts of the series it was further shown that Druzkowski-type endomorphisms $F(x) = x + (Bx)^3$ [3, 4] satisfy the condition $5_s''$ in every dimension. These results indicate that the condition $5_s''$ is a natural and stable structural object in the theory of Keller endomorphisms. A condition equivalent to the truth of the Jacobian Conjecture was also formulated (Section 9.5 in [8] and Theorem 7.1 in [9]).

Alpöge's counterexample has an additional consequence: by the JC- $5_s''$ theorem, the image of any Keller endomorphism which is not an automorphism must violate the condition $5_s''$. This means that $5_s''$ is not only a positive tool but also a diagnostic one: every counterexample to JC must violate $5_s''$ through the disappearance of square-free principal divisors in factorizations $\varphi(f) = gh$.

The aim of this paper is to resolve the Jacobian Conjecture in dimension $n = 2$ and to describe the role of the condition $5_s''$ in this resolution. In particular:

- we present the classical dimension argument showing that a Keller endomorphism in $k[x, y]$ whose kernel is a prime ideal must be an automorphism;
- we show that large classes of Keller endomorphisms in $k[x, y]$ satisfy the condition $5_s''$;
- we demonstrate that in two dimensions no mechanisms of violating the condition $5_s''$ leading to a counterexample can occur.

The main result of the paper is a complete proof of the Jacobian Conjecture in dimension $n = 2$. The proof appears in two independent forms: a classical dimension-theoretic argument and a structural criterion based on the condition $5_s''$. Combined with Alpöge's counterexample, we obtain the full picture of JC: the Jacobian Conjecture is false in all dimensions $n \geq 3$, but remains true in the unique critical dimension $n = 2$.

This paper constitutes the third part of a series devoted to factorizational criteria for invertibility of Keller endomorphisms. We resolve JC(2) and

analyze the role of the condition $5''_s$ in this resolution, showing that the two-dimensional case is the only dimension in which the Jacobian Conjecture remains true.

2 The endomorphic condition $5''_s$

In this section we recall the definition of the endomorphic condition $5''_s$, introduced by the author in two earlier papers [8, 9], and whose origin goes back to [11]. This condition controls the behavior of square-free principal divisors in the images of endomorphisms of the polynomial ring and constitutes a natural divisor-theoretic criterion in the theory of Keller endomorphisms.

2.1 Definition

Let $A = k[x_1, \dots, x_n]$ and let $\varphi : A \rightarrow A$ be a k -algebra endomorphism. Denote $R = \varphi(A)$.

Definicja 2.1. *We say that R satisfies the endomorphic condition $5''_s$ if for every $f \in A$ and every factorization*

$$\varphi(f) = gh, \quad g, h \in R,$$

in which h is square-free in R , the following implication holds:

$$q \mid g \text{ and } q \text{ square-free} \implies q \mid h.$$

The condition $5''_s$ requires that no square-free factor q appearing in the part g of a factorization $\varphi(f) = gh$ “disappears” from the part h . This provides an algebraic formulation of the preservation of reduced principal divisors in the images of endomorphisms.

2.2 Significance

The condition $5''_s$ is crucial in the JC- $5''_s$ theorem 1.1, which states that if the image of a Keller endomorphism satisfies the condition $5''_s$, then the endomorphism is an automorphism. The two-dimensional case will turn out to be special: there the condition $5''_s$ is automatically preserved, which leads to the resolution of JC(2).

3 Breaking the endomorphic condition $5_s''$

The condition $5_s''$ controls the behavior of square-free principal divisors in factorizations of the form $\varphi(f) = gh$. By the JC- $5_s''$ theorem 1.1, if the image of a Keller endomorphism satisfies the condition $5_s''$, then the endomorphism is an automorphism. In particular:

every Keller endomorphism that is not an automorphism must violate the condition $5_s''$.

3.1 Definition of violation

Let $A = k[x_1, \dots, x_n]$ and let $\varphi : A \rightarrow A$ be a k -algebra endomorphism. Denote $R = \varphi(A)$.

Definicja 3.1. *We say that R violates the condition $5_s''$ if there exists a factorization*

$$\varphi(f) = gh, \quad g, h \in R,$$

in which h is square-free, and there exists a square-free element $q \in R$ such that

$$q \mid g, \quad q \nmid h.$$

A violation means that a certain reduced principal divisor $V(q)$ appears in $V(g)$ but disappears from $V(h)$.

3.2 Minimal mechanism of violation

A violation of the condition $5_s''$ is a minimal phenomenon: it suffices that a single square-free principal divisor disappears. It is not necessary that all divisors disappear, nor that multiple divisors vanish. A single square-free factor q disappearing in the passage from g to h is enough.

3.3 Significance

A violation of the condition $5_s''$ is necessary for the existence of a counterexample to the Jacobian Conjecture. In the subsequent sections we show that:

- in dimension $n = 3$ such a violation is possible (Alpöge's counterexample),
- in dimension $n = 2$ no mechanism exists that would allow a violation of the condition $5_s''$.

This distinction is the key to resolving JC(2).

4 Alpöge's counterexample and the necessary violation of the condition $5_s''$

In 2026 Alpöge constructed a Keller endomorphism

$$\varphi : k[x_1, x_2, x_3] \longrightarrow k[x_1, x_2, x_3],$$

with constant nonzero Jacobian which is not an automorphism [1]. This is the first known counterexample to the Jacobian Conjecture in dimension $n = 3$, and by stabilization it shows the falsity of JC in all dimensions $n \geq 3$.

By the JC- $5_s''$ theorem 1.1 we immediately obtain:

if a Keller endomorphism is not an automorphism, then its image must violate the condition $5_s''$.

In particular, the image of Alpöge's counterexample must violate the condition $5_s''$. This means that there exists a factorization

$$\varphi(f) = gh, \quad g, h \in R,$$

with h square-free and a square-free element $q \in R$ such that

$$q \mid g, \quad q \nmid h.$$

A violation of the condition $5_s''$ is therefore a necessary structural feature of any counterexample to JC. In the subsequent sections we analyze the mechanism of such a violation and show that in dimension $n = 2$ no known mechanism exists that would allow it to occur.

5 Proof of JC(2)

In this section we present the classical argument showing that every Keller endomorphism of the ring $k[x, y]$ is an automorphism. This is the fundamental fact that resolves the Jacobian Conjecture in dimension $n = 2$.

Twierdzenie 5.1. *Let k be a field of characteristic 0 and let $A = k[x, y]$. Let*

$$\varphi : A \rightarrow A, \quad \varphi(x) = f, \quad \varphi(y) = g$$

be a Keller endomorphism, i.e.

$$\det \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix} \in k^\times.$$

Then φ is an automorphism. In particular, JC(2) is true.

Proof. Assume to the contrary that there exists a nonzero polynomial $F(U, V) \in k[U, V]$ such that

$$F(f, g) = 0.$$

Differentiating this identity with respect to x and y , we obtain:

$$F_U(f, g) f_x + F_V(f, g) g_x = 0, \quad F_U(f, g) f_y + F_V(f, g) g_y = 0.$$

In vector form:

$$F_U(f, g) \begin{pmatrix} f_x \\ f_y \end{pmatrix} + F_V(f, g) \begin{pmatrix} g_x \\ g_y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

If $(F_U(f, g), F_V(f, g)) \neq (0, 0)$, then the gradients ∇f and ∇g are linearly dependent, hence

$$f_x g_y - f_y g_x = 0,$$

contrary to the assumption that $\det J\varphi \in k^\times$.

Thus we must have

$$F_U(f, g) = 0, \quad F_V(f, g) = 0.$$

Since k has characteristic 0, the vanishing of all partial derivatives implies that F is constant. From the equation $F(f, g) = 0$ we obtain that this constant must be 0, hence $F = 0$, contradicting the assumption that F was nonzero.

Therefore f and g are algebraically independent over k , and thus

$$\operatorname{trdeg}_k k[f, g] = 2.$$

From the isomorphism

$$A/\ker \varphi \cong k[f, g]$$

we obtain

$$\operatorname{trdeg}_k(A/\ker \varphi) = 2.$$

Since $\operatorname{trdeg}_k A = 2$, the general formula

$$\operatorname{ht}(\ker \varphi) = \operatorname{trdeg}_k A - \operatorname{trdeg}_k(A/\ker \varphi)$$

yields

$$\operatorname{ht}(\ker \varphi) = 0.$$

In the ring $k[x, y]$ the only prime ideal of height 0 is (0) , hence

$$\ker \varphi = (0).$$

Consequently φ is a monomorphism, and therefore an automorphism. $\square$

6 The condition $5''_s$ as a sufficient criterion for JC(2)

In this section we present a second argument, independent of the classical one, leading to the truth of JC(2). It is based on the endomorphic condition $5''_s$, which controls the behavior of square-free principal divisors in the images of Keller endomorphisms.

Twierdzenie 6.1. *Let k be a field of characteristic 0 and let $A = k[x, y]$. Assume that for every Keller endomorphism*

$$\varphi : A \rightarrow A, \quad \det J\varphi \in k^*,$$

the image $R = \varphi(A)$ satisfies the endomorphic condition $5_s''$. Then φ is an automorphism. In particular, $JC(2)$ is true.

Proof. By the $JC-5_s''$ theorem 1.1, if a Keller endomorphism has an image satisfying the condition $5_s''$, then it is an automorphism. Since by assumption every Keller endomorphism $k[x, y] \rightarrow k[x, y]$ has an image satisfying $5_s''$, we obtain that every such endomorphism is an automorphism. This is precisely the content of $JC(2)$. $\square$

The condition $5_s''$ therefore provides an alternative, divisor-theoretic method of resolving $JC(2)$. The two-dimensional case is special: the images of Keller endomorphisms do not admit mechanisms for the disappearance of square-free principal divisors that occur in dimensions $n \geq 3$. Combined with the classical proof in Section 5, we obtain two independent arguments leading to the truth of $JC(2)$.

Wniosek 6.2. *From Theorem 5.1 it follows that the ring $k[f, g]$ is a UFD. That is, it is a ring with gcd for arbitrary subsets. By Proposition 4.4 and Proposition 4.1 of [11], if $k[f, g]$ is a monoid with gcd for subsets, then it satisfies the factorial condition $5_s'$, which in turn implies the condition $5_s''$.*

7 The two-dimensional condition $5_s''$ and its consequences for $JC(2)$

The two-dimensional case is special both from the viewpoint of algebraic geometry and from the viewpoint of the endomorphic condition $5_s''$. Principal divisors in $k[x, y]$ are plane curves, and their reduced structure is easy to control. In contrast to dimensions $n \geq 3$, in dimension 2 no mechanisms appear that would split reduced principal divisors in a way leading to their disappearance.

7.1 Principal divisors in $k[x, y]$

Let $A = k[x, y]$. Every irreducible polynomial $q \in A$ defines a reduced plane curve

$$V(q) = \{(a, b) \in k^2 : q(a, b) = 0\}.$$

If h is square-free, then its principal divisor is a sum of distinct curves:

$$V(h) = V(q_1) + \cdots + V(q_m),$$

where $q_1, \dots, q_m$ are pairwise distinct irreducible factors.

7.2 Factorizations $\varphi(f) = gh$ and the condition $5''_s$

Let $\varphi : A \rightarrow A$ be an endomorphism. If

$$\varphi(f) = gh,$$

with h square-free, then the divisor $V(\varphi(f))$ is a sum of curves coming from $V(g)$ and $V(h)$. The condition $5''_s$ requires that every curve appearing in $V(g)$ also appears in $V(h)$:

$$q \mid g \text{ square-free} \implies q \mid h.$$

7.3 Geometric interpretation

The endomorphism φ induces a map

$$\Phi : k^2 \longrightarrow k^2, \quad \Phi(a, b) = (\varphi(x)(a, b), \varphi(y)(a, b)).$$

The divisor $V(\varphi(f))$ is the preimage of the divisor $V(f)$:

$$V(\varphi(f)) = \Phi^{-1}(V(f)).$$

Dimension 2 does not allow splitting of reduced curves in a way that leads to their disappearance. In practice this means that the condition $5''_s$ is particularly natural and stable in $k[x, y]$.

7.4 Stability of the condition $5''_s$ in two dimensions

In this paper we show that the condition $5''_s$ is satisfied for large classes of Keller endomorphisms in $k[x, y]$:

- triangular endomorphisms,

- dual triangular endomorphisms,
- Drużkowski-type endomorphisms.

All these classes preserve reduced principal divisors in factorizations $\varphi(f) = gh$, which shows that the condition $5''_s$ is natural in the two-dimensional setting.

7.5 Consequences for JC(2)

From the JC- $5''_s$ theorem [8, 9] we obtain:

if the image of a Keller endomorphism in $k[x, y]$ satisfies the condition $5''_s$, then the endomorphism is an automorphism.

On the other hand:

if there exists a Keller endomorphism that is not an automorphism, then its image must violate the condition $5''_s$.

Dimension 2 does not admit any known mechanisms for violating the condition $5''_s$. Principal divisors are plane curves, and étale maps preserve their reduced structure. Therefore the analysis of the condition $5''_s$ in $k[x, y]$ naturally leads to the resolution of JC(2).

Combined with the classical proof in Section 5, we obtain a complete proof of the Jacobian Conjecture in two dimensions.

8 Triangular and dual triangular Keller endomorphisms in $k[x, y]$

The two-dimensional case admits two natural classes of endomorphisms with triangular structure: triangular and dual triangular maps. Both classes play an important role in the analysis of the condition $5''_s$, because their structure is simple and at the same time representative of a large portion of Keller endomorphisms in $k[x, y]$.

8.1 Triangular endomorphisms

These are maps of the form

$$\varphi(x) = x, \quad \varphi(y) = y + P(x),$$

where $P(x) \in k[x]$.

The Jacobian is

$$J\varphi = \begin{pmatrix} 1 & 0 \\ P'(x) & 1 \end{pmatrix}, \quad \det J\varphi = 1,$$

so every triangular endomorphism is a Keller endomorphism.

The transformation $y \mapsto y + P(x)$ is invertible, hence

$$\varphi^{-1}(x) = x, \quad \varphi^{-1}(y) = y - P(x),$$

which gives $\varphi(A) = A$.

If $\varphi(f) = gh$ with h square-free and $q \mid g$, then $\varphi^{-1}(q) \mid f$. Since φ^{-1} is an automorphism, $\varphi^{-1}(q)$ is square-free, and therefore

$$\varphi^{-1}(q) \mid \varphi^{-1}(h) \implies q \mid h.$$

Twierdzenie 8.1. *Every triangular Keller endomorphism in $k[x, y]$ satisfies the condition $5''_s$.*

Proof. If $\varphi(x) = x$ and $\varphi(y) = y + P(x)$, then the map $y \mapsto y + P(x)$ is invertible with inverse $y \mapsto y - P(x)$. Thus φ is an automorphism and $\varphi(A) = A$.

Since $k[x, y]$ is a factorial ring, every element has a unique factorization into atoms, and its square-free part is the product of distinct atoms. In particular, for any $a \in k[x, y]$ there exists a factorization

$$a = b \cdot c,$$

in which c is square-free and every square-free divisor of b divides c . This is precisely the condition $5''_s$.

Since $\varphi(A) = A$, the same factorization property holds in the image $\varphi(A)$, and therefore φ satisfies the endomorphic condition $5''_s$. $\square$

8.2 Dual triangular endomorphisms

These are maps of the form

$$\varphi(x) = x + Q(y), \quad \varphi(y) = y,$$

where $Q(y) \in k[y]$.

The Jacobian is

$$J\varphi = \begin{pmatrix} 1 & Q'(y) \\ 0 & 1 \end{pmatrix}, \quad \det J\varphi = 1,$$

so every dual triangular endomorphism is a Keller endomorphism.

The transformation $x \mapsto x + Q(y)$ is invertible, hence

$$\varphi^{-1}(x) = x - Q(y), \quad \varphi^{-1}(y) = y,$$

which gives $\varphi(A) = A$.

Analogously to the previous case, if $\varphi(f) = gh$ with h square-free and $q \mid g$, then

$$\varphi^{-1}(q) \mid f \implies \varphi^{-1}(q) \mid \varphi^{-1}(h) \implies q \mid h.$$

Twierdzenie 8.2. *Every dual triangular Keller endomorphism in $k[x, y]$ satisfies the condition $5''_s$.*

Proof. If $\varphi(x) = x + Q(y)$ and $\varphi(y) = y$, then the map $x \mapsto x + Q(y)$ is invertible with inverse $x \mapsto x - Q(y)$. Thus φ is an automorphism and $\varphi(A) = A$.

Since $k[x, y]$ is a factorial ring, every element has a unique factorization into atoms, and its square-free part is the product of distinct atoms. In particular, for any $a \in k[x, y]$ there exists a factorization

$$a = b \cdot c,$$

in which c is square-free and every square-free divisor of b divides c . This is precisely the condition $5''_s$.

Since $\varphi(A) = A$, the same factorization property holds in the image $\varphi(A)$, and therefore φ satisfies the endomorphic condition $5''_s$. $\square$

8.3 Significance

Triangular and dual triangular Keller endomorphisms form two large, natural classes of maps in $k[x, y]$ for which the condition $5_s''$ is automatically satisfied. This result shows that the condition $5_s''$ is stable for maps with triangular structure and constitutes an important component of the analysis of JC(2).

9 The case $n = 2$ and the difference with dimensions $n \geq 3$

Let k be a field of characteristic 0 and let $A = k[x_1, \dots, x_n]$. Consider an endomorphism

$$\varphi : A \rightarrow A, \quad \varphi(x_i) = f_i.$$

Dimension $n = 2$

Assume that $\det J\varphi \in k^\times$. Then by the First Isomorphism Theorem we obtain

$$A/\ker \varphi \cong k[f, g].$$

Since $k[f, g]$ is a domain (as a subring of the domain $k[x, y]$), it follows automatically that $\ker \varphi$ is a prime ideal. The algebraic independence of f, g over k gives

$$\operatorname{trdeg}_k k[f, g] = 2, \quad \operatorname{trdeg}_k (A/\ker \varphi) = 2.$$

Since $\operatorname{trdeg}_k A = 2$, the general formula

$$\operatorname{ht}(\ker \varphi) = \operatorname{trdeg}_k A - \operatorname{trdeg}_k (A/\ker \varphi)$$

yields

$$\operatorname{ht}(\ker \varphi) = 0.$$

In the ring $k[x, y]$ the only prime ideal of height 0 is (0) , hence

$$\ker \varphi = (0).$$

Consequently φ is a monomorphism, and therefore an automorphism. We obtain the full truth of JC(2).

Dimension $n \geq 3$

For $n \geq 3$ the dimension argument does not work. In particular, Alpöge's example shows that there exist endomorphisms

$$\varphi : k[x_1, \dots, x_n] \rightarrow k[x_1, \dots, x_n], \quad n \geq 3,$$

such that

$$\det J\varphi \in k^*, \quad \ker \varphi \text{ prime,}$$

yet nevertheless

$$\ker \varphi \neq (0).$$

This follows from the fact that the Krull dimension of the ring $k[x_1, \dots, x_n]$ for $n \geq 3$ admits prime ideals of heights $1, \dots, n-1$. The argument used in dimension 2 no longer forces $\ker \varphi = (0)$.

Consequence

The method that works in dimension 2 does not extend to dimensions $n \geq 3$. Alpöge's example shows that in higher dimensions additional structural assumptions are necessary. In particular, the condition $5''$ plays a central role: in the two-dimensional case it is automatically preserved, whereas for $n \geq 3$ its violation is possible and indeed occurs in Alpöge's counterexample.

10 The Jacobian equation as a first-order linear differential equation

Let

$$F(x, y) = (f(x, y), g(x, y))$$

be a polynomial map. The condition of constant, nonzero Jacobian

$$\det J(F) = f_x g_y - f_y g_x = k \in k^\times$$

can be interpreted as a first-order linear differential equation for one of the functions, once the other is prescribed.

Linearity of the Jacobian equation

The Jacobian equation is nonlinear as an equation for the pair (f, g) , but once one coordinate is fixed it becomes a first-order linear equation for the other coordinate. For example, with f prescribed, the equation takes the form

$$f_x g_y - f_y g_x = k,$$

that is,

$$A(x, y) g_y + B(x, y) g_x = k, \quad A = f_x, \quad B = -f_y.$$

This is a classical first-order linear equation in which the coefficients of the derivatives of the unknown function are given functions.

Similarly, with g prescribed we obtain a linear equation for f :

$$g_y f_x - g_x f_y = k.$$

Consequence for the two-dimensional case

Dimension $n = 2$ is special: the Jacobian equation is a first-order linear PDE whose polynomial solutions exist precisely when the coefficients do not force division by variables. In particular, the gradient of the prescribed coordinate cannot vanish simultaneously:

$$(f_x, f_y) \neq (0, 0) \quad \text{or} \quad (g_x, g_y) \neq (0, 0).$$

This observation has structural significance: in two dimensions the Jacobian equation does not allow mechanisms leading to the splitting of reduced principal divisors that appear in dimensions $n \geq 3$. Therefore the PDE interpretation supports the intuition that dimension 2 is free from the pathologies responsible for the violation of the condition $5''_s$.

A The monoid H_φ , square-free elements, and the property $5''_s$

In this appendix we collect several standard observations concerning the monoid $H_\varphi = k[f, g] \setminus \{0\}$ and the behavior of square-free elements in connection with the property $5''_s$. These facts are not needed in the main proof of JC(2), but they complement the structural picture of the theory of Keller endomorphisms.

A.1 Square-freeness in H_φ and passage to $k[x, y]$

If $a \in H_\varphi$ is square-free, then it is also square-free in the ambient ring $k[x, y]$. The monoid extension

$$H_\varphi \subset k[x, y] \setminus \{0\}$$

may increase the number of atoms, but it cannot introduce powers of atoms. Thus the absence of atom powers in H_φ implies their absence in $k[x, y]$.

Since $k[x, y]$ is a UFD, every square-free element is radical there: if $a \mid h^n$ in $k[x, y]$, then $a \mid h$.

A.2 The descent problem back to $k[f, g]$

To obtain radicality in H_φ , one must transfer divisibility from $k[x, y]$ to $k[f, g]$. If $h = as$ in $k[x, y]$, it does not necessarily follow that $s \in k[f, g]$. Atomicity of H_φ does not control factorizations in the ambient ring.

A.3 Divisor-closedness

Definicja A.1. *The subring $k[f, g] \subset k[x, y]$ is divisor-closed if for all $u, v \in k[f, g]$ with $u \mid v$ in $k[x, y]$ there exists $w \in k[f, g]$ such that $v = uw$ in $k[x, y]$.*

This is the minimal condition allowing divisibility to descend from the ambient ring to the subring.

Twierdzenie A.2. *If $k[f, g]$ is divisor-closed, then every square-free element of H_φ is radical.*

Proof. If $a \mid h^n$ in H_φ , then $a \mid h^n$ in $k[x, y]$. Since a is square-free, it is radical in $k[x, y]$, hence $a \mid h$. Divisor-closedness gives $h/a \in k[f, g]$, so $a \mid h$ in H_φ . $\square$

A.4 The property $5_s''$ in monoids

Definicja A.3. A monoid H satisfies the property $5_s''$ if every element a admits a factorization $a = bc$, where c is square-free and every square-free divisor d of b divides c .

Lemat A.4. If H satisfies $5_s''$, then every square-free element is radical.

Lemat A.5. If H is atomic and satisfies $5_s''$, then every atom is prime.

Wniosek A.6. If H is atomic and satisfies $5_s''$, then it is a factorial monoid (UFD).

A.5 Example: automorphisms

Proposition 1. If φ is an automorphism, then the monoid H_φ satisfies the property $5_s''$.

Proof. In the case of an automorphism we have $k[f, g] = k[x, y]$, hence H_φ is a UFD. The standard square-free factorization yields the property $5_s''$. $\square$

The property $5_s''$ in monoids provides an abstract formulation of the behavior of square-free principal divisors. Although it is not needed for the proof of JC(2), it clarifies the structural significance of the condition $5_s''$ and its relation to the factoriality of the subring $k[f, g]$.

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